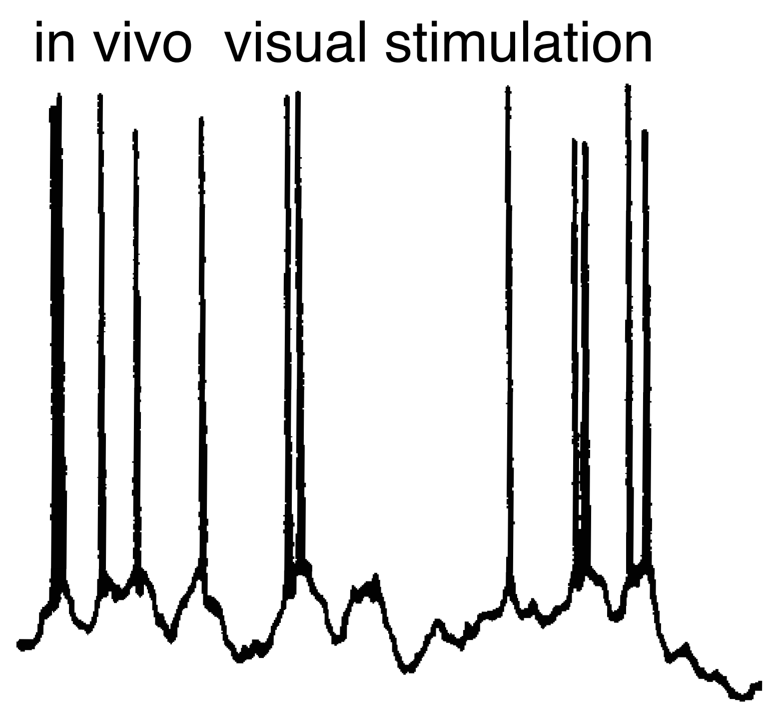
The Neural Code

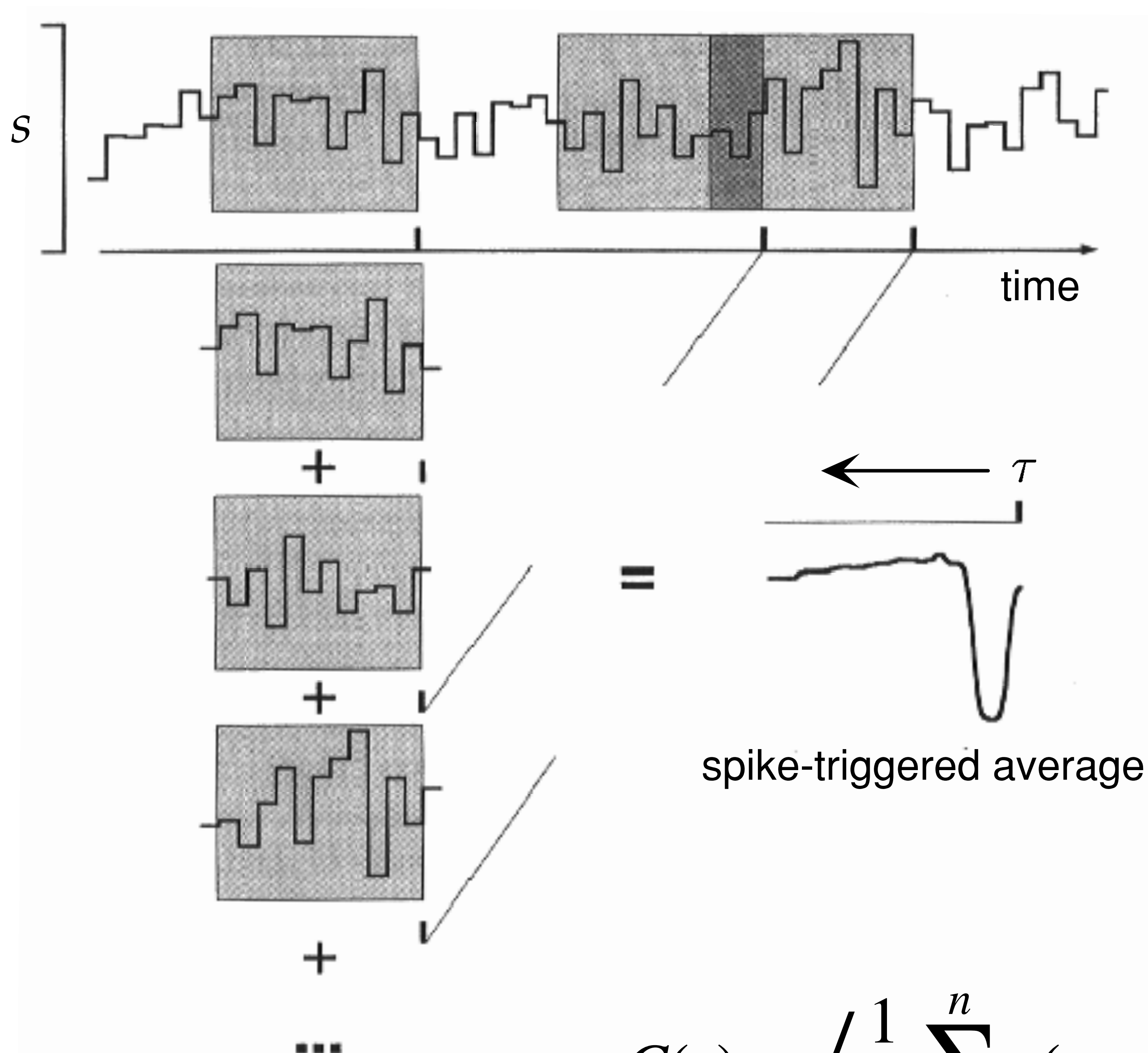
Quan Wen Nov 25

Spike trains convey information



$$\rho(t) = \sum_i \delta(t - t_i)$$

What kind of information a spike train represent?



$$C(\tau) = \left\langle \frac{1}{n} \sum_{i=1}^{n} s(t_i - \tau) \right\rangle$$

Linear response theory and optimal kernel (whiteboard)

Temporal receptive field

$$D_t(\tau) = \alpha \exp(-\alpha\tau)\left[\frac{(\alpha\tau)^5}{5!} - \frac{(\alpha\tau)^7}{7!}\right]$$

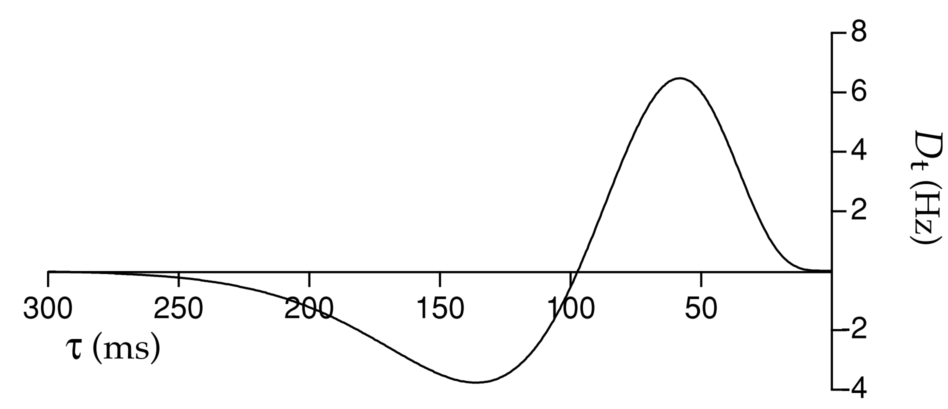
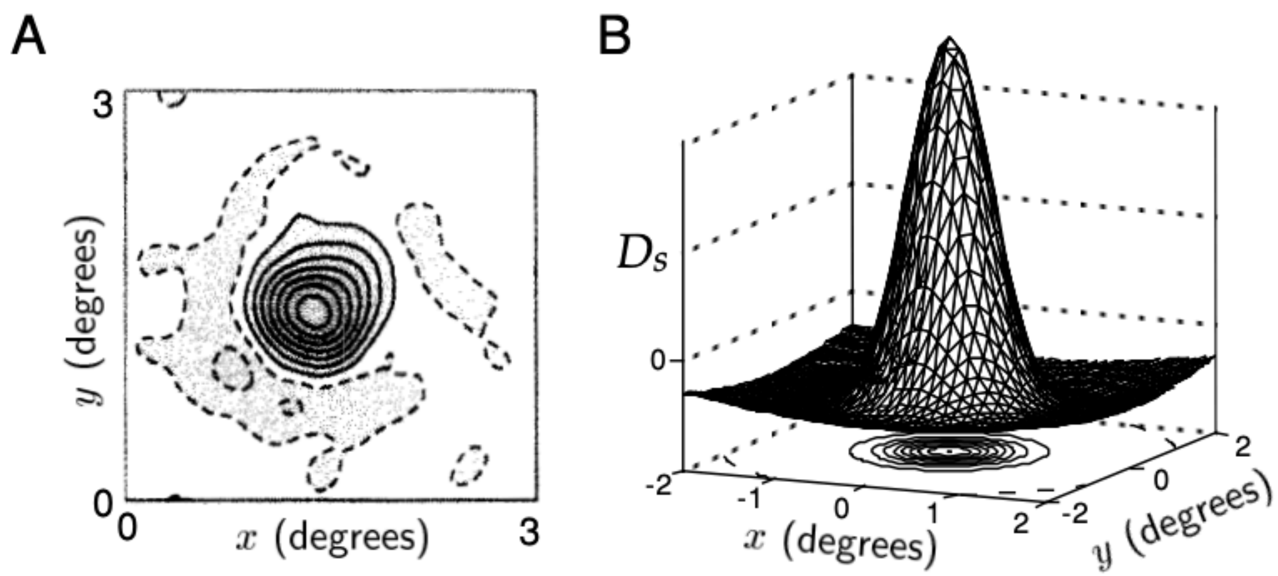


Figure 2.14 Temporal structure of a receptive field. The function $D_t(\tau)$ of equation 2.29 with $\alpha = 1/(15\text{ ms})$.

Spatial receptive field of retinal ganglion cells



Differences between Gaussians

$$D_s(x, y) = \pm\left(\frac{1}{2\pi\sigma_{cen}^2}\exp\left(-\frac{x^2+y^2}{2\sigma_{cen}^2}\right) - \frac{B}{2\pi\sigma_{sur}^2}\exp\left(-\frac{x^2+y^2}{2\sigma_{sur}^2}\right)\right).$$

Receptive field of simple cells in the visual cortex

$$D_s(x,y) = \frac{1}{2\pi\sigma_x\sigma_y}\exp\Big(-\frac{x^2}{2\sigma_x^2}-\frac{y^2}{2\sigma_y^2}\Big)\cos(kx-\phi)$$

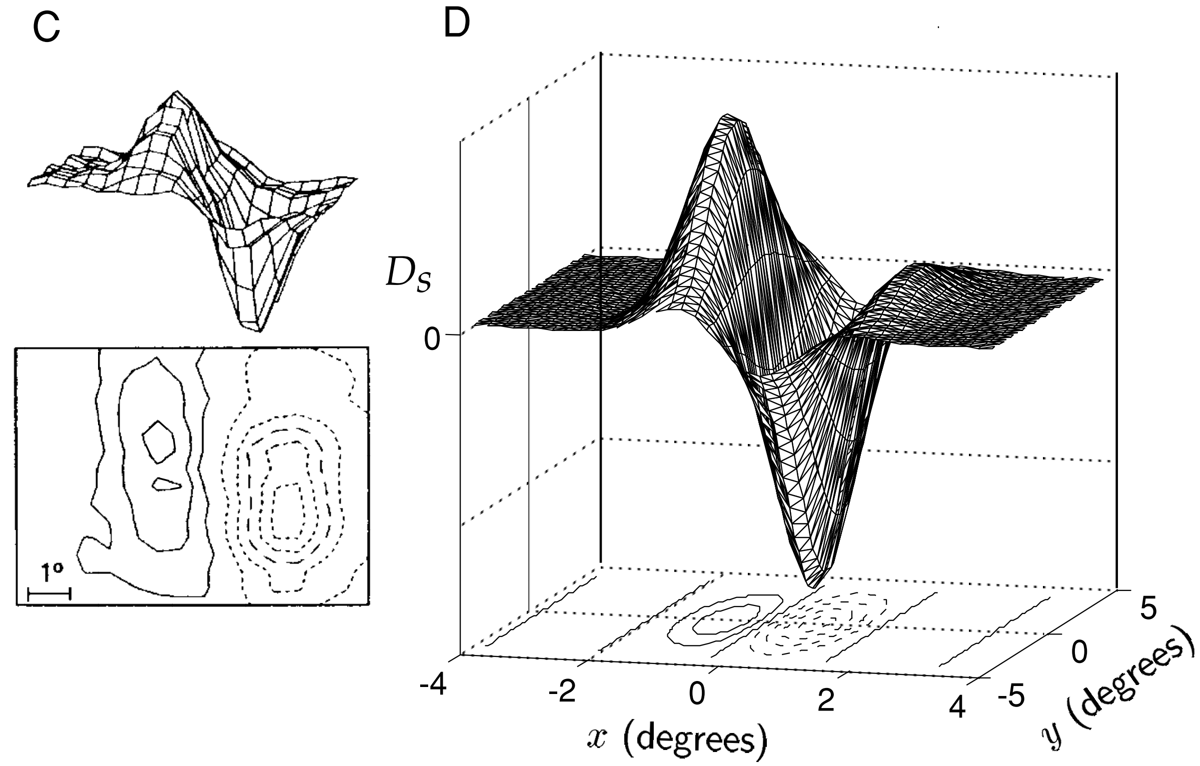
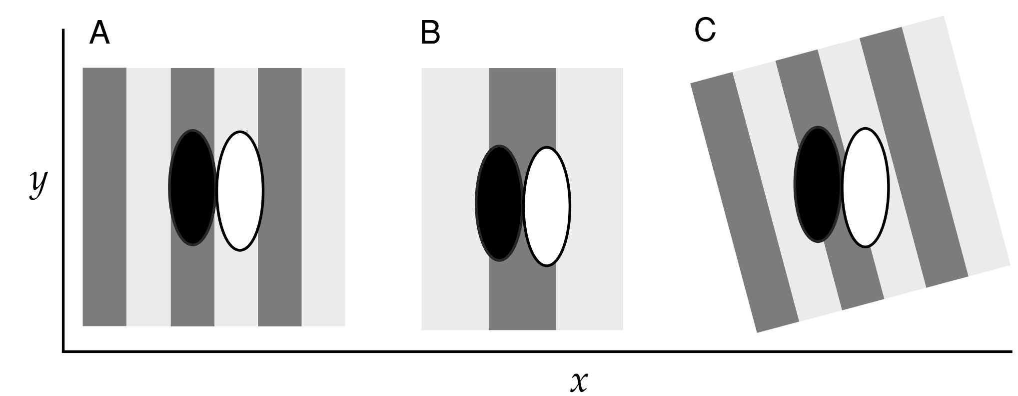


Figure 2.10 Spatial receptive field structure of simple cells. (A) and (C) Spatial structure of the receptive fields of two neurons in cat primary visual cortex determined by averaging stimuli between 50 ms and 100 ms prior to an action potential. The upper plots are three-dimensional representations, with the horizontal dimensions acting as the x-y plane and the vertical dimension indicating the magnitude and sign of $D_s(x,y)$. The lower contour plots represent the x-y plane. Regions with solid contour curves are ON areas where $D_s(x,y) > 0$, and regions with dashed contours are OFF areas where $D_s(x,y) < 0$. (B) and (D) Gabor functions (equation 2.27) with $\sigma_x = 1^\circ$, $\sigma_y = 2^\circ$, $1/k = 0.56^\circ$, and $\phi = 1 - \pi/2$ (B) or $\phi = 1 - \pi$ (D), chosen to match the receptive fields in A and C. (A and C adapted from Jones and Palmer, 1987a.)



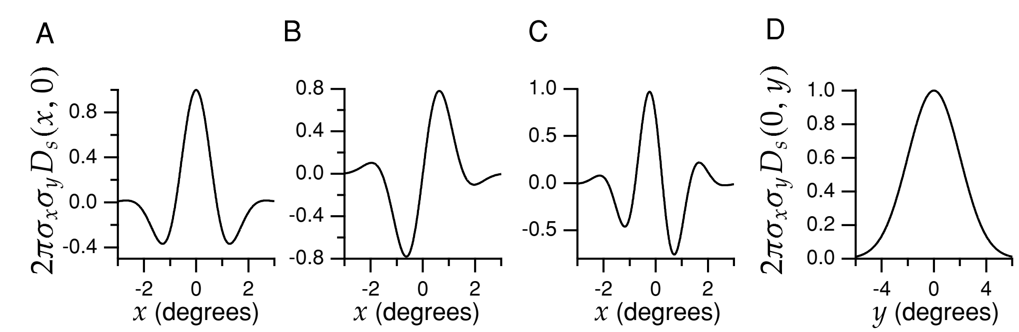
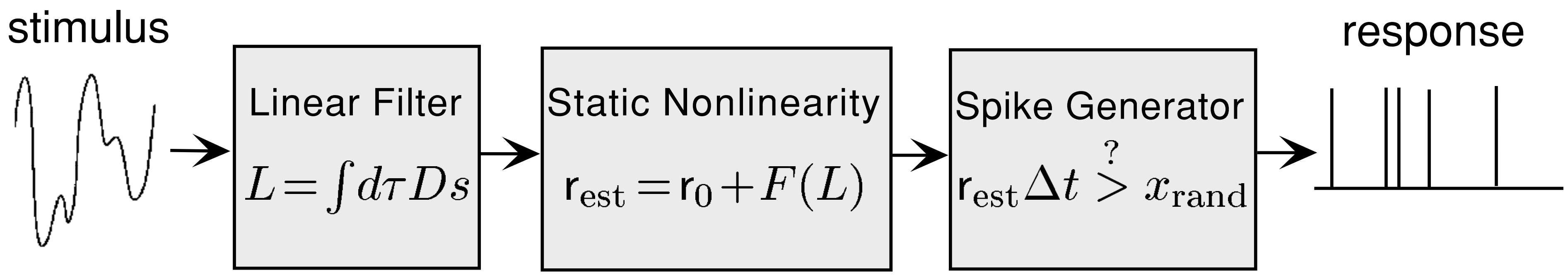
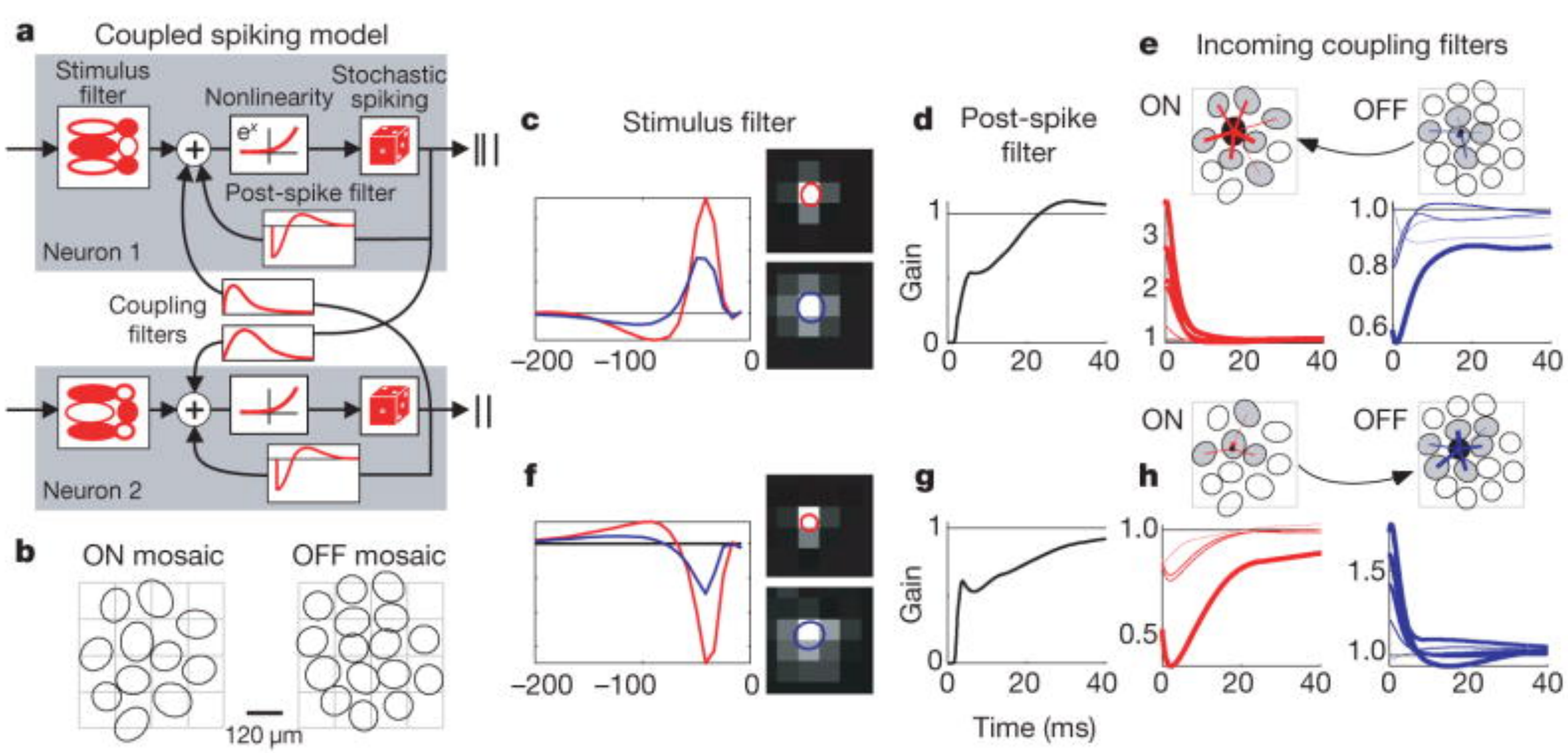


Figure 2.12 Gabor functions of the form given by equation 2.27. For convenience we plot the dimensionless function $2\pi\sigma_x\sigma_y D_s$. (A) A Gabor function with $\sigma_x = 1^\circ$, $1/k = 0.5^\circ$, and $\phi = 0$ plotted as a function of x for $y = 0$. This function is symmetric about $x = 0$. (B) A Gabor function with $\sigma_x = 1^\circ$, $1/k = 0.5^\circ$, and $\phi = \pi/2$ plotted as a function of x for $y = 0$. This function is antisymmetric about $x = 0$ and corresponds to using a sine instead of a cosine function in equation 2.27. (C) A Gabor function with $\sigma_x = 1^\circ$, $1/k = 0.33^\circ$, and $\phi = \pi/4$ plotted as a function of x for $y = 0$. This function has no particular symmetry properties with respect to $x = 0$. (D) The Gabor function of equation 2.27 with $\sigma_y = 2^\circ$ plotted as a function of y for $x = 0$. This function is simply a Gaussian.

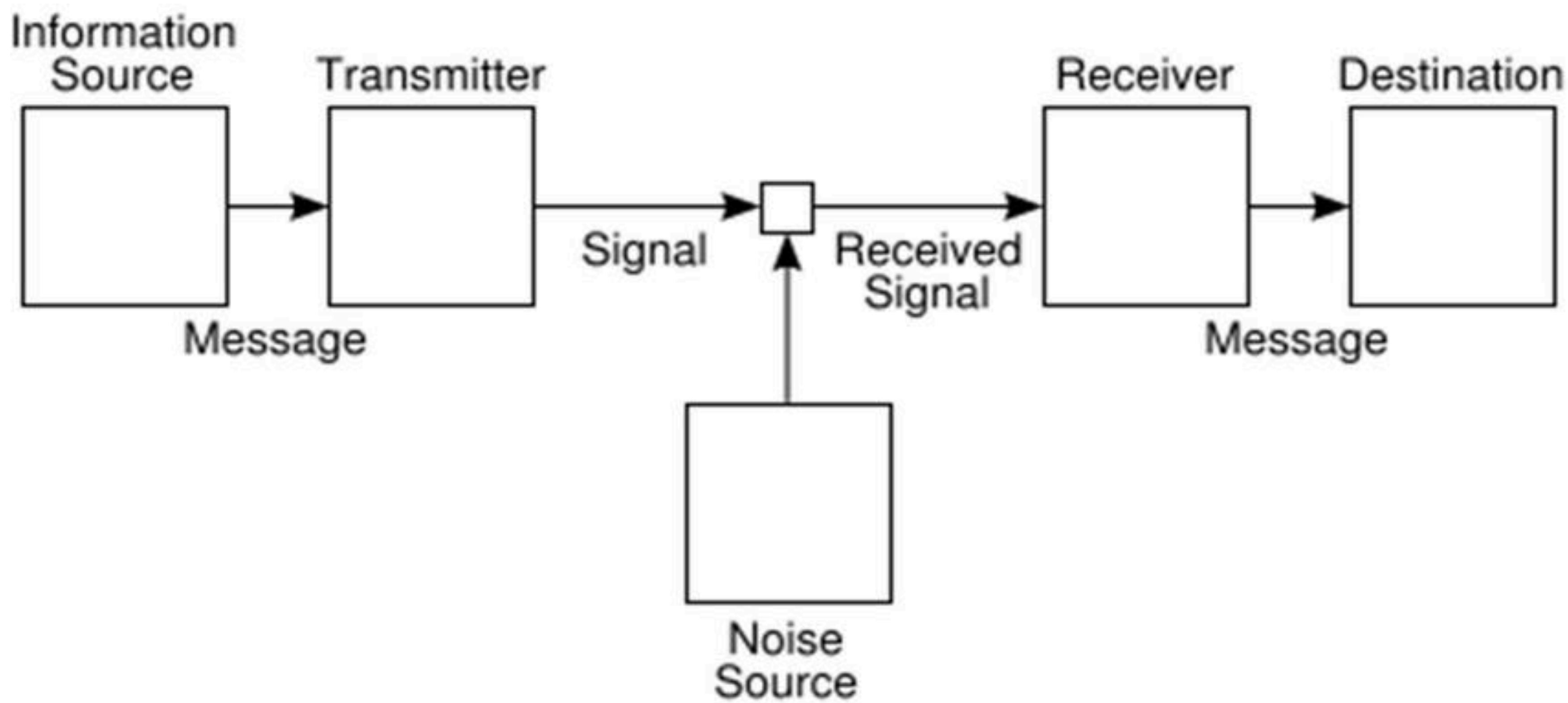
Linear-nonlinear Model





Pillow J et al Nature 2009

Information Theory



Shannon Information

- Assumption 1: if all *N* possible answers are equally likely, then the information gained should be a monotonically increasing function of *N.*
- Assumption 2: if our question consists of two *independent* two parts, then the total information gained as the sum of the information gained in response to each of the two subquestions.
- Assumption 3: Multipart questions can be viewed as branching tree structures.

Shannon Information

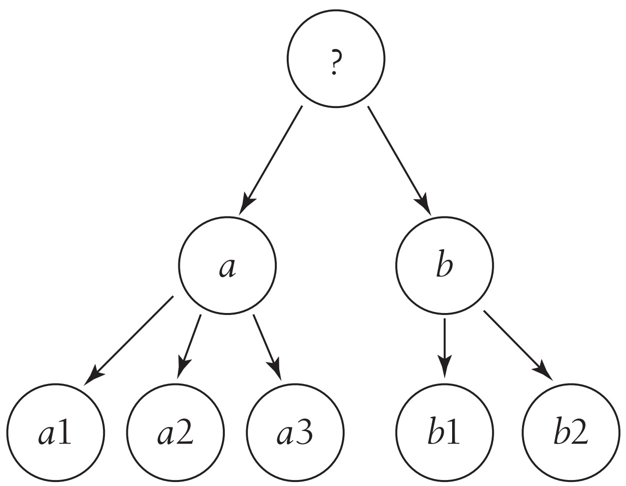


FIGURE 6.1

The branching postulate in Shannon's proof. The idea is to break a big question into multiple parts, as in the familiar game of twenty questions. We start with some initial question, at the top (?). Depending on the answer to this question (*a* or *b*), we ask a new question. This second question in turn has multiple possible answers (*a*1, *a*2, *a*3 or *b*1, *b*2). In this tree structure, the various subquestions live at branch points, with the answers emerging along the branches; finding our way to the full answer means following one path through the tree. The average information that we gain along this path should be additive—the weighted sum of information gained at every branch point.

William Bialek Biophysics, Searching for Principles

The game of 63

What's the smallest number of yes/no questions needed to identify an integer *x* between 0 and 63?

- $x \geq 32$?
- is $x \bmod 32 \geq 16$?
- is $x \bmod 16 \geq 8$?
- is $x \bmod 8 \geq 4$?
- is $x \bmod 4 \geq 2$?
- is $x \bmod 2 \geq 1$?

$I = \log_2 64 = 6$ bit

Shannon Information

$$h = \log_2 N \text{ when } P(r) = \frac{1}{N}$$

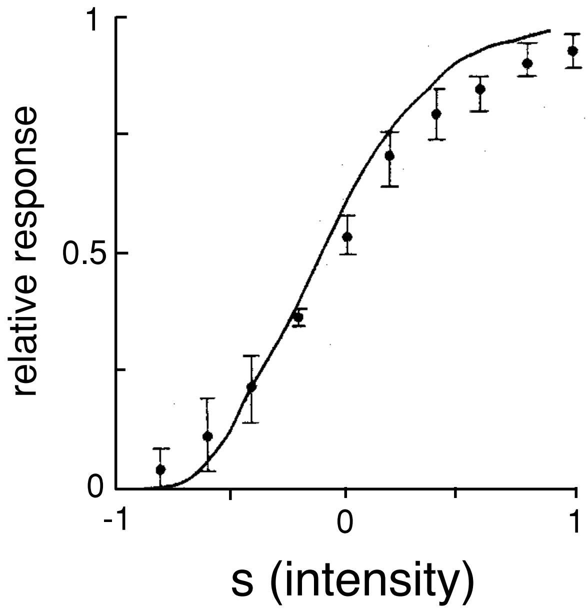
$$I = -\sum_r P(r) \log_2 P(r)$$

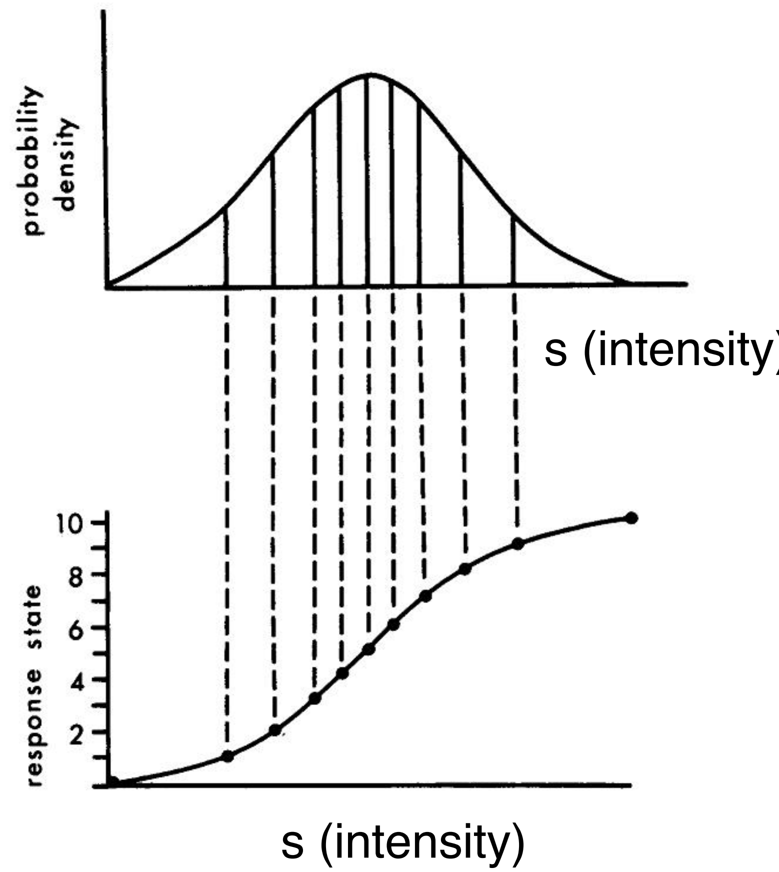
Continuous limit

$$I = -\sum_r p(r)\Delta r \log_2 p(r)\Delta r$$

$$\lim_{\Delta r \to 0} (I + \log_2 \Delta r) = -\int dr p(r) \log_2 p(r)$$

Maximizing entropy principle



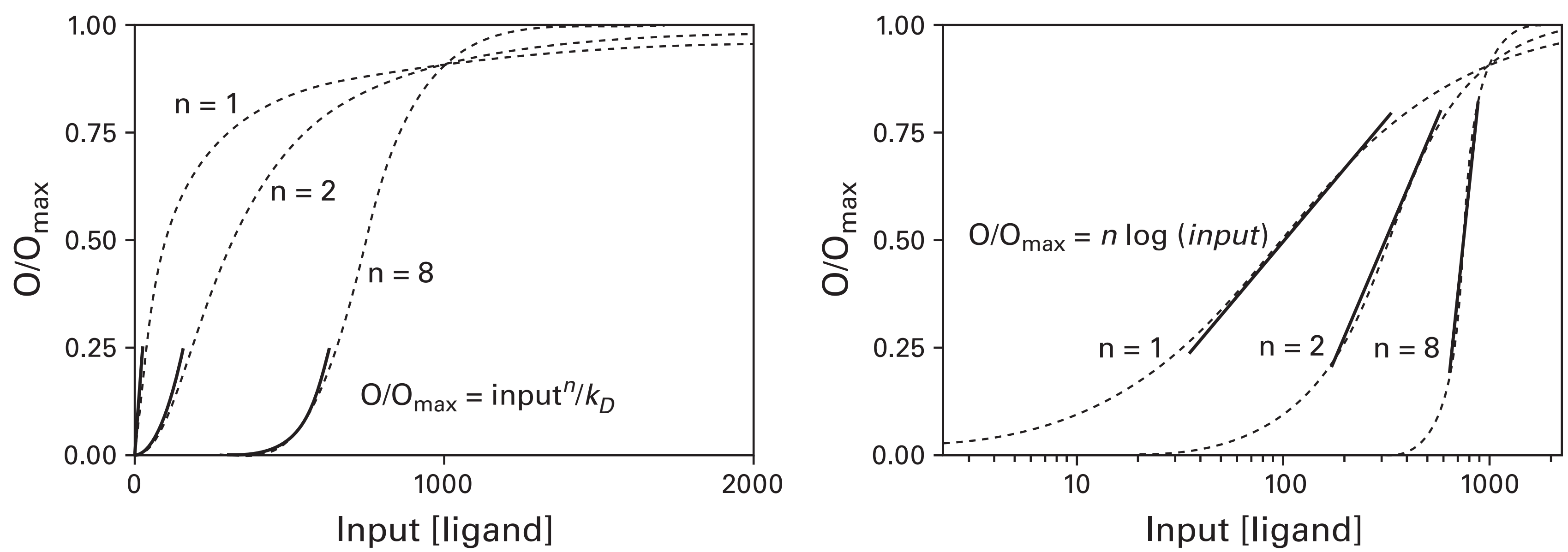


$$r = f(s)$$

$$H = -\int_0^{r_{max}} dr p(r) \ln p(r) \qquad \frac{df}{ds} = r_{max} p(s)$$

$$p(r) = \frac{1}{r_{max}}$$

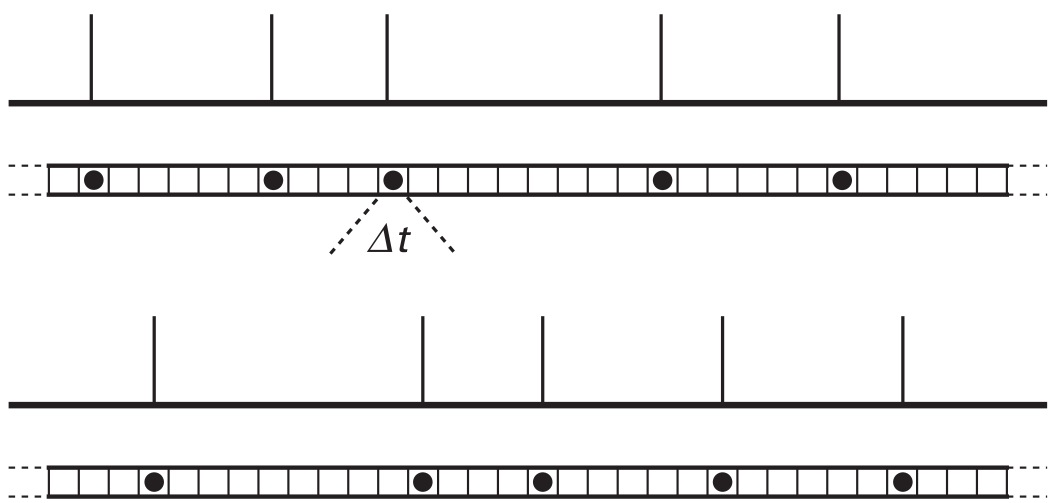
I/O function to match input distribution



$$\frac{O}{O_{max}} = \frac{[ligand]^n}{k_D + [ligand]^n}$$

n : number of binding sites

Information and entropy



$$T = \frac{1\text{ sec}}{\Delta t}$$

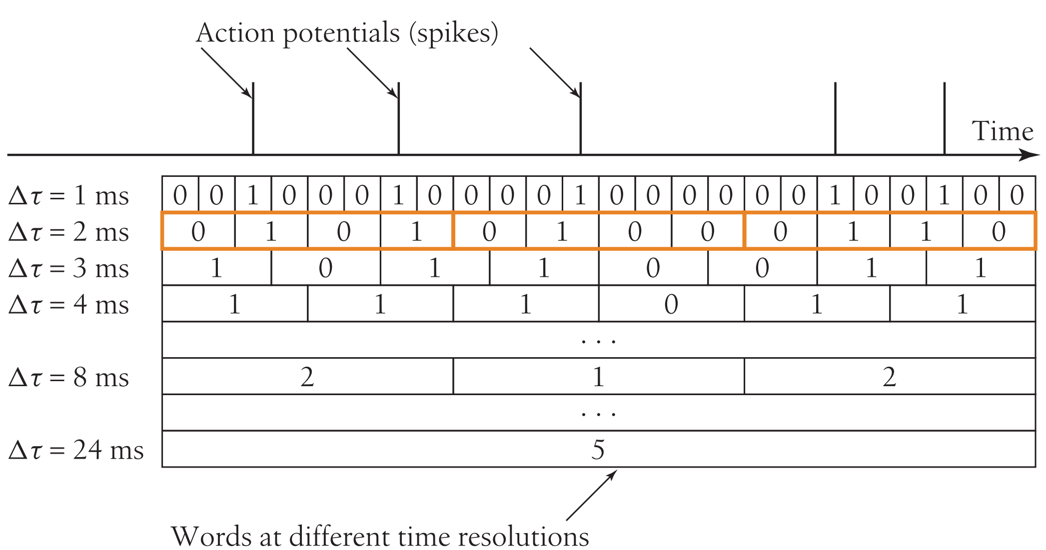
$$M = \frac{T!}{R!(T-R)!}$$

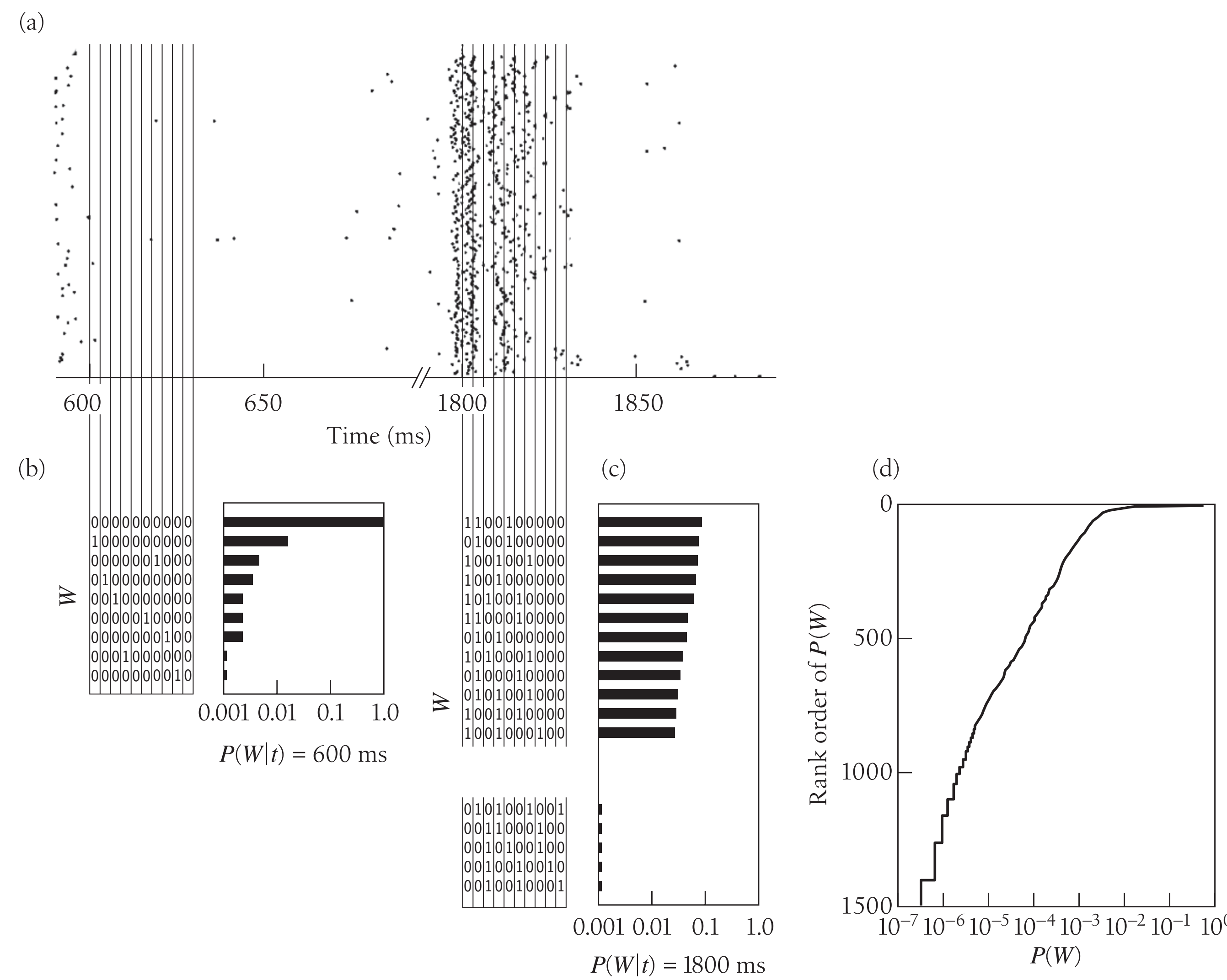
$$I = \log M \text{ (bit)}$$

Quantities to measure

- Bits per second (information rate)
- Bits per spike (Does every spike convey information?)
- Bits per cost (No pain, no gain)

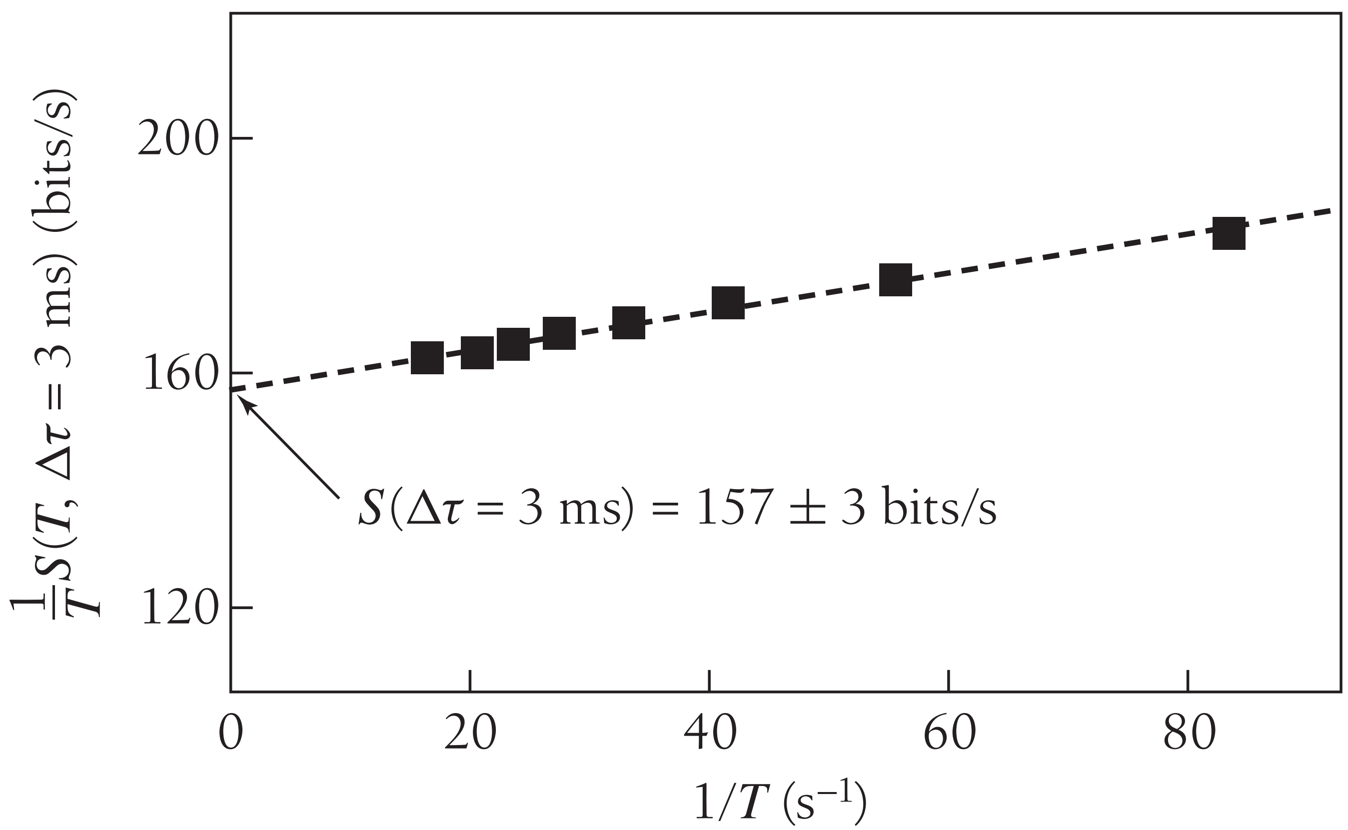
Information rate (spikes)





De Ruyter van Steveninck et al (1997)

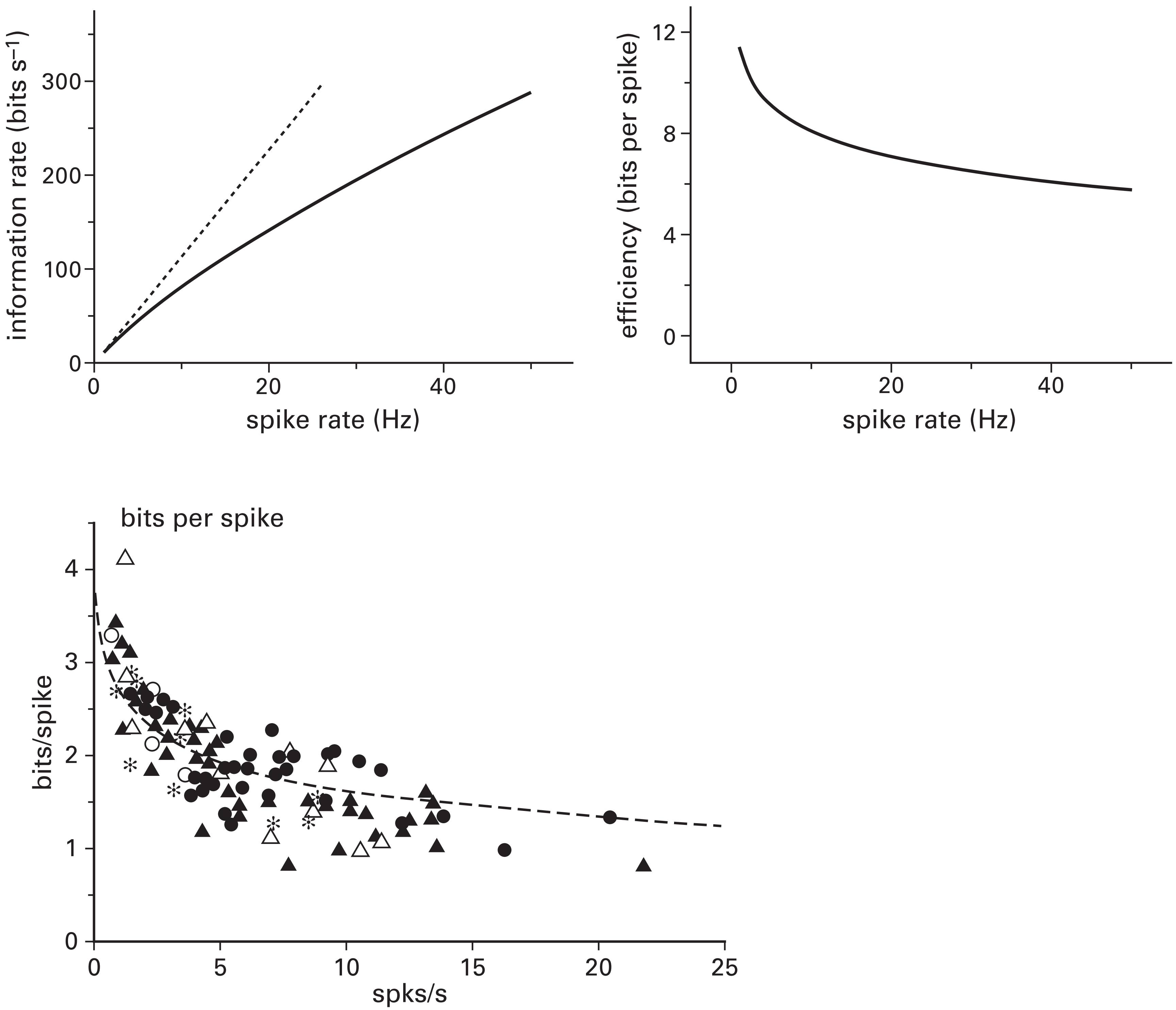
Information rate (spikes)



Strong et al (1998)

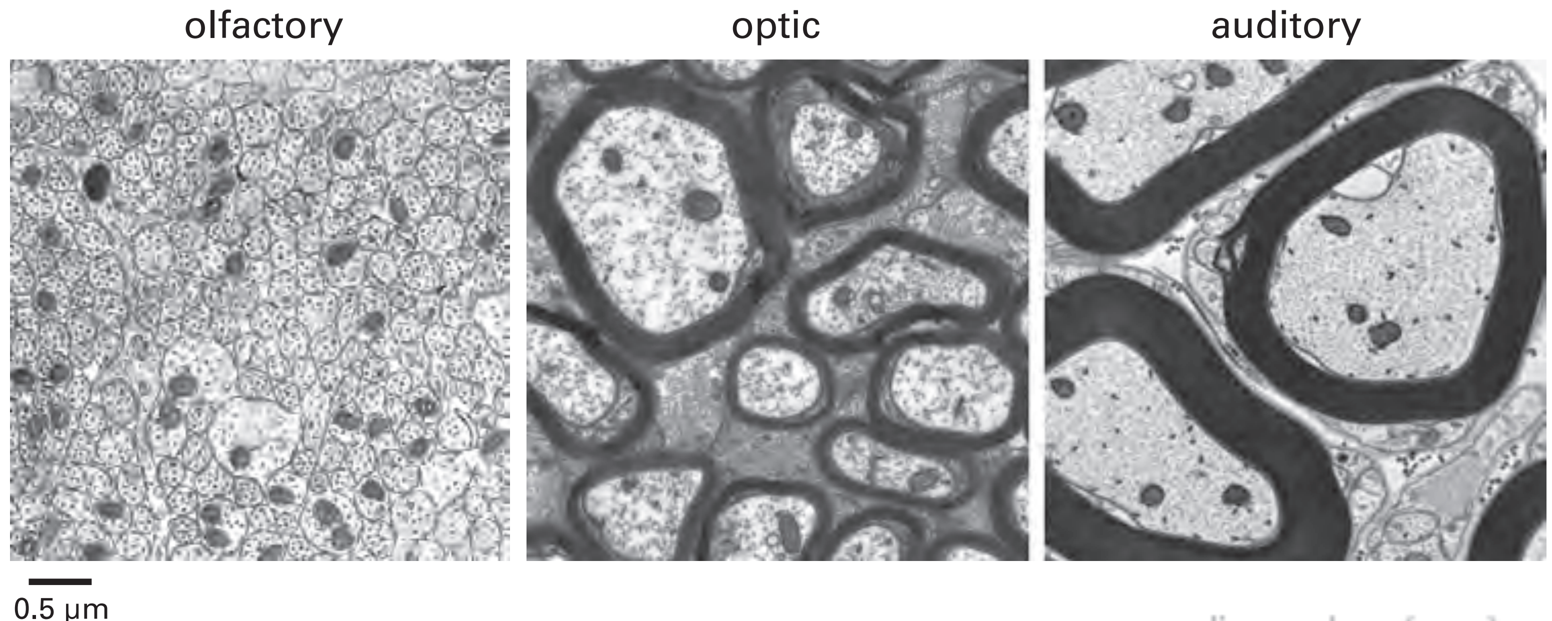
$$\frac{1}{T}S(T, \Delta\tau) = \mathcal{S}(\Delta\tau) + \frac{C(\Delta\tau)}{T} + \cdots,$$

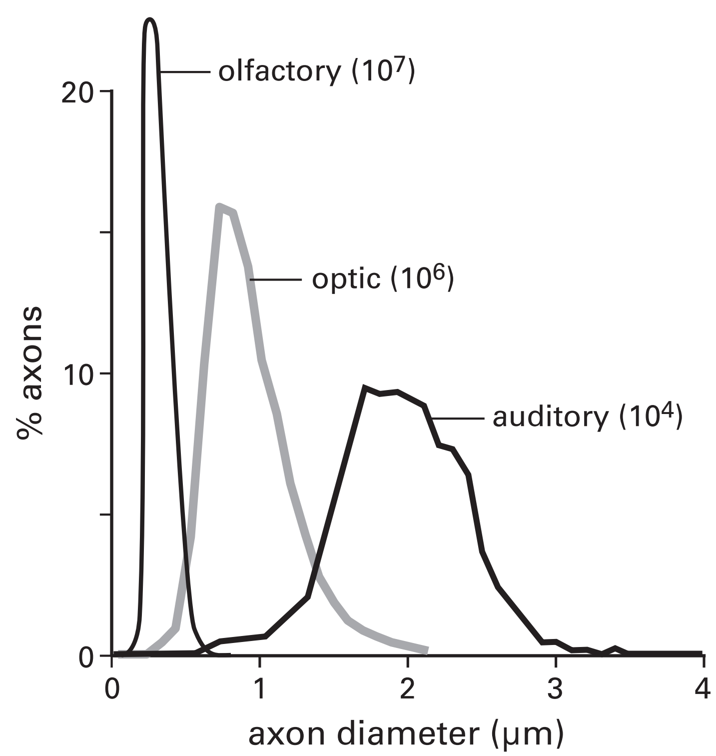
Does every spike count?

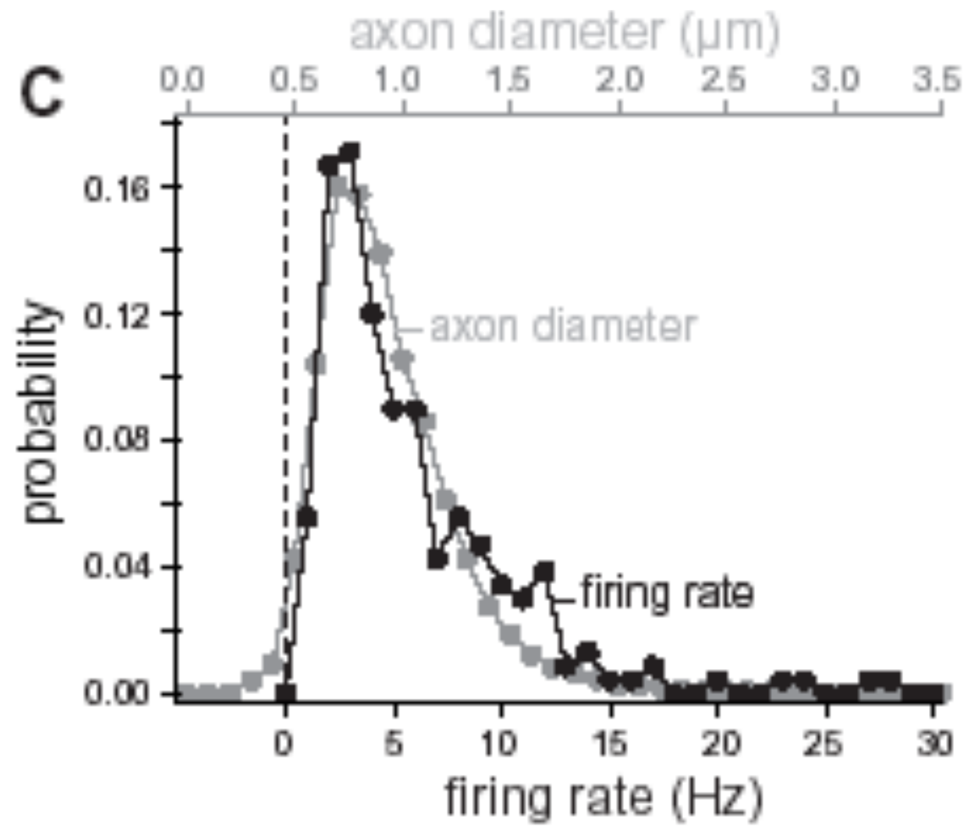


Balasubramanian & Sterling 2009

Information transmission is costly

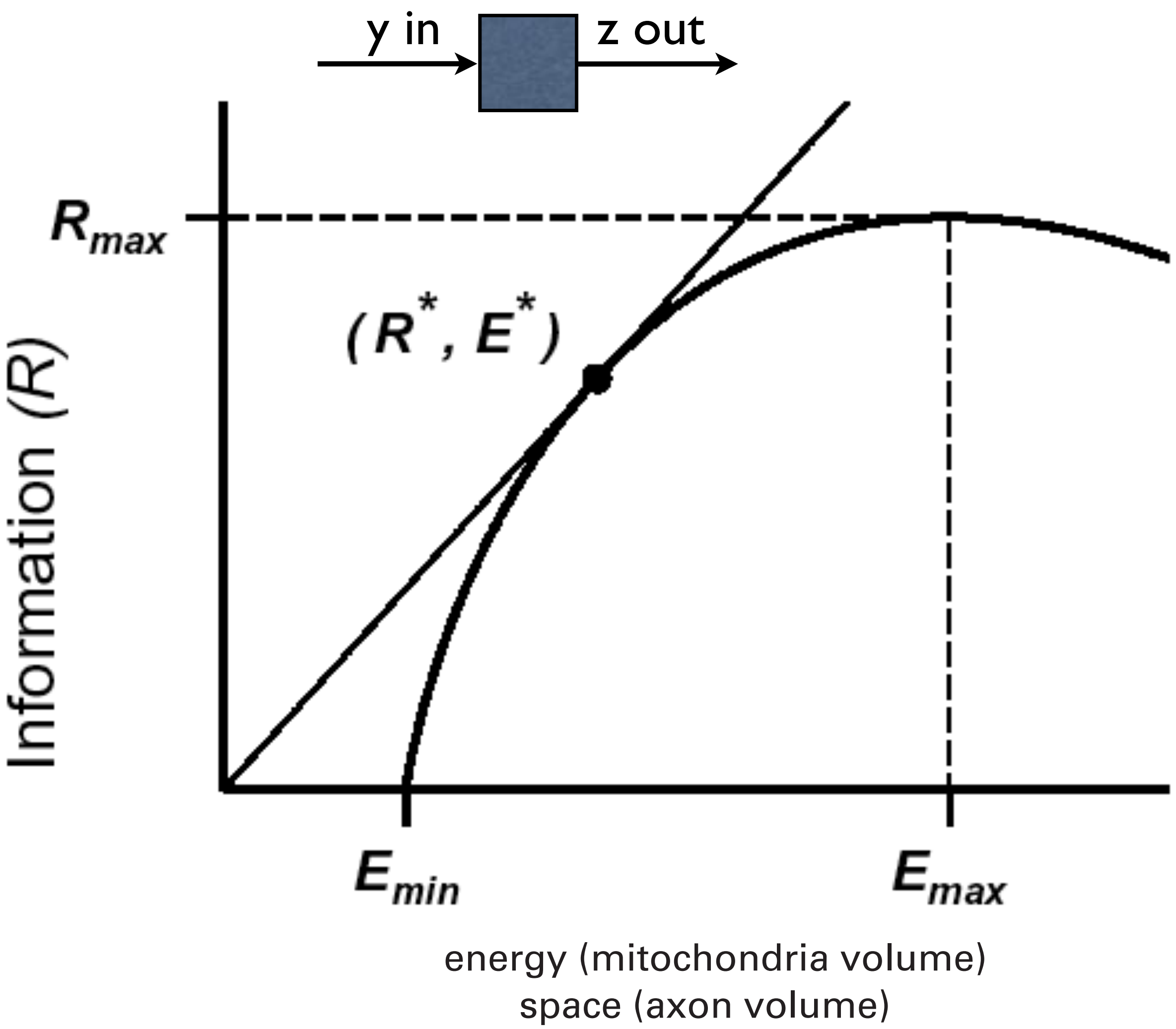


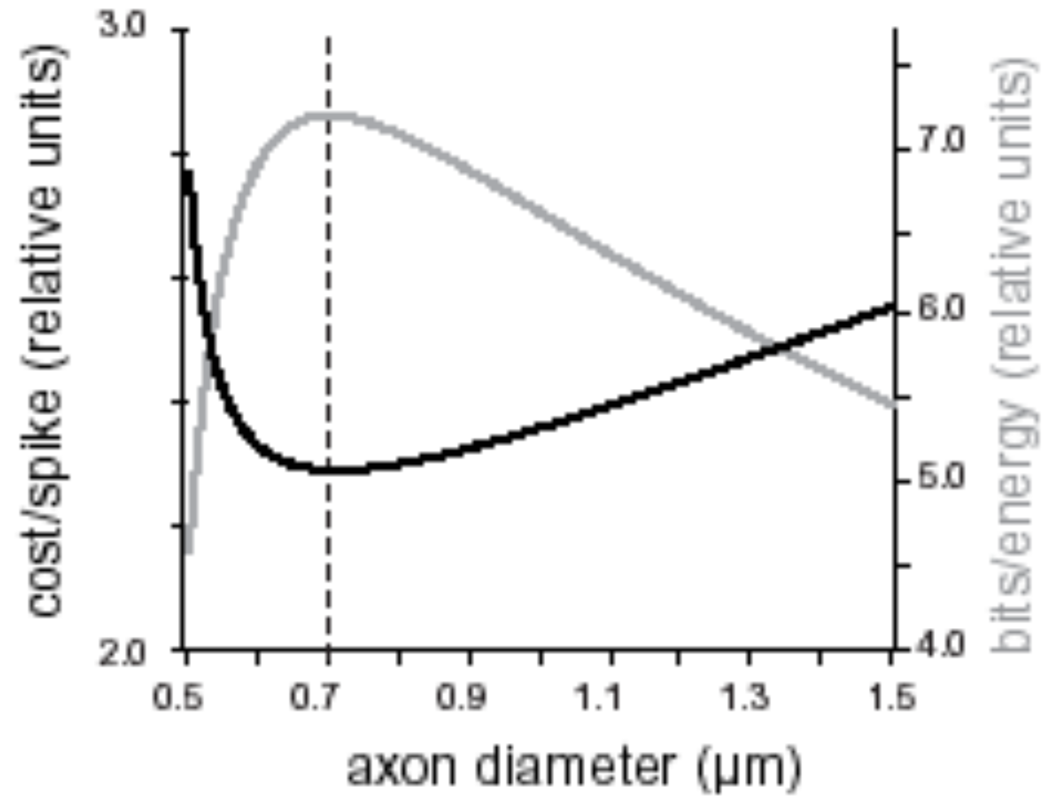




Perge, J. A., Niven, J. E., Mugnaini, E., Balasubramanian, V., & Sterling, P. (2012). *Journal of Neuroscience*, *32*, 626–638.

Law of diminished returns





Heterogeneity & efficiency in the brain Vijay Balasubramanian 2015

Perge et al, 2018 How the optic nerve allocates space, energy capacity, and information

Mutual Information (whiteboard)

Information rate (analog signal with noise)

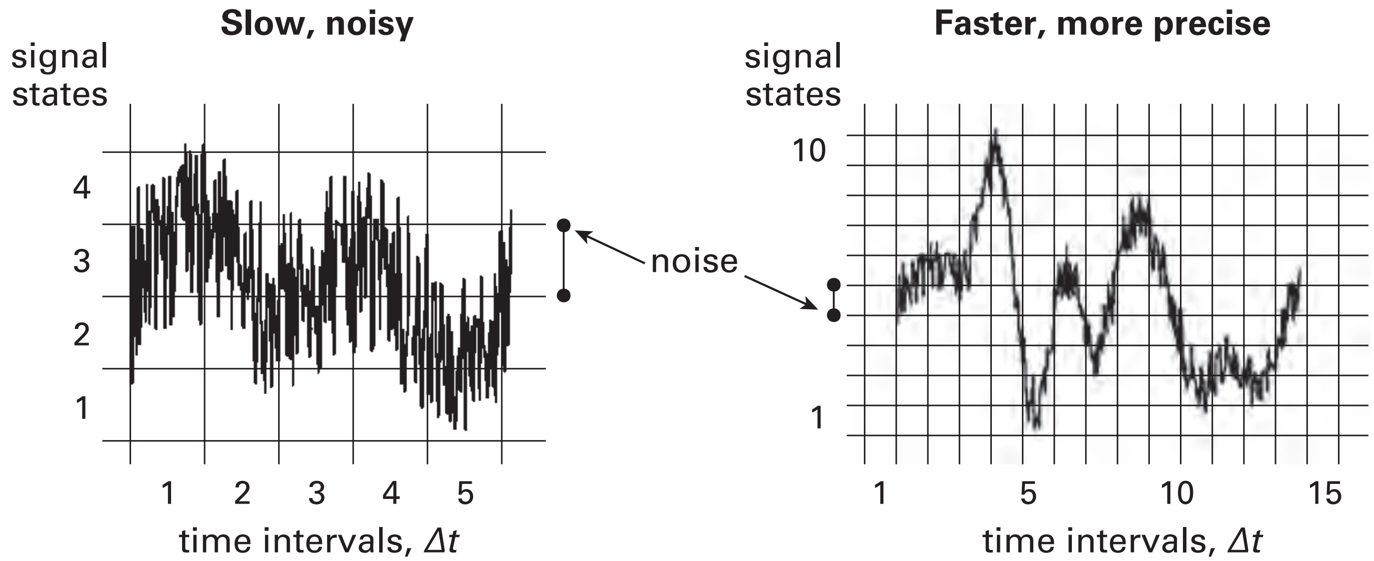


Figure 5.3

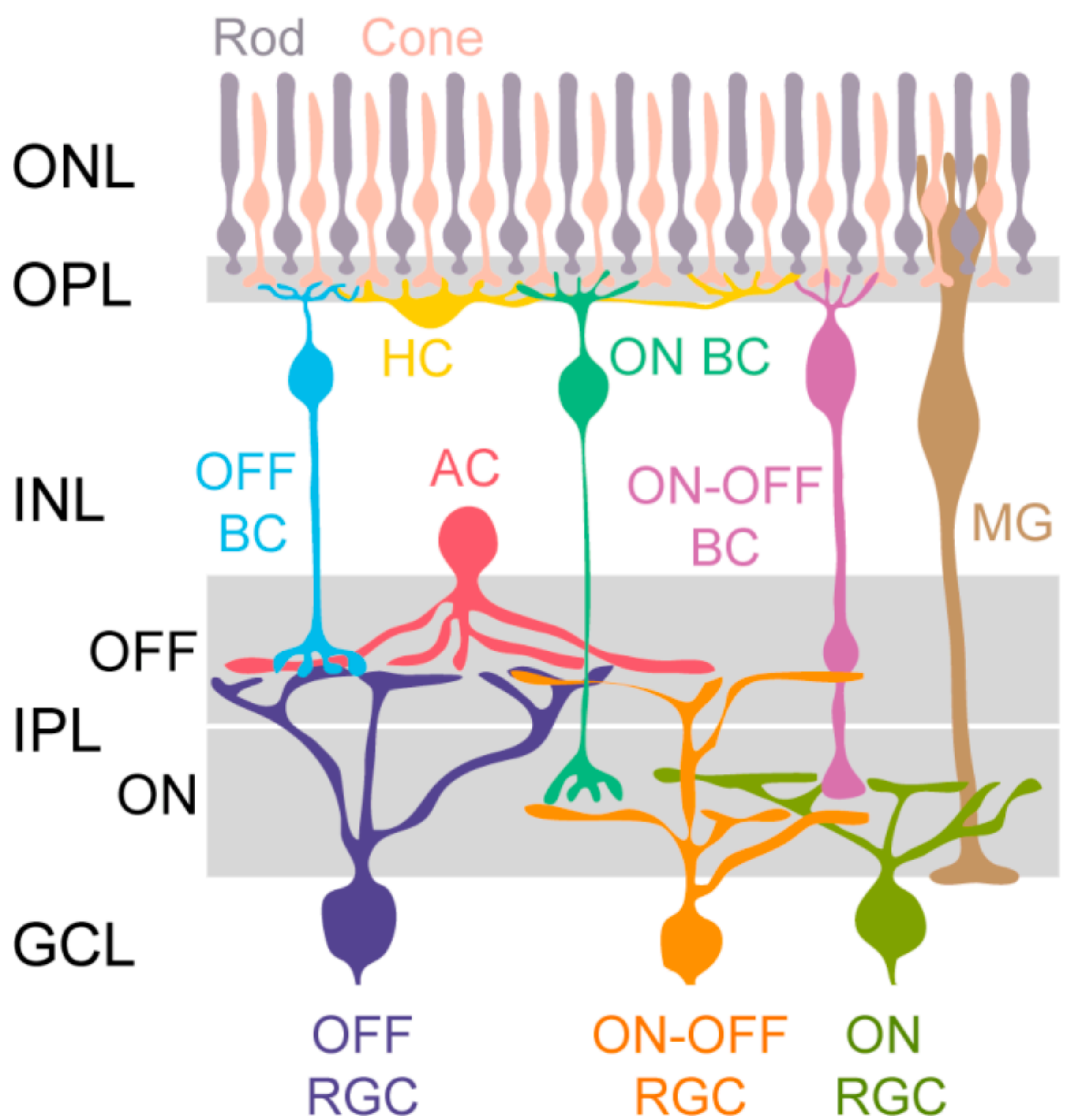
Signal range, noise, and response dynamics determine the information rates of analogue signals. Noise divides a waveform's signal range into discriminable states, and states can change at time intervals Δt. The faster, more reliable waveform obviously conveys more details of the signal. From equation 5.5, it also has a higher information rate because it has a higher S/N and, with shorter Δt, changes level at a higher rate.

$$I = \frac{1}{\Delta t} \log_2(1 + S/N)$$

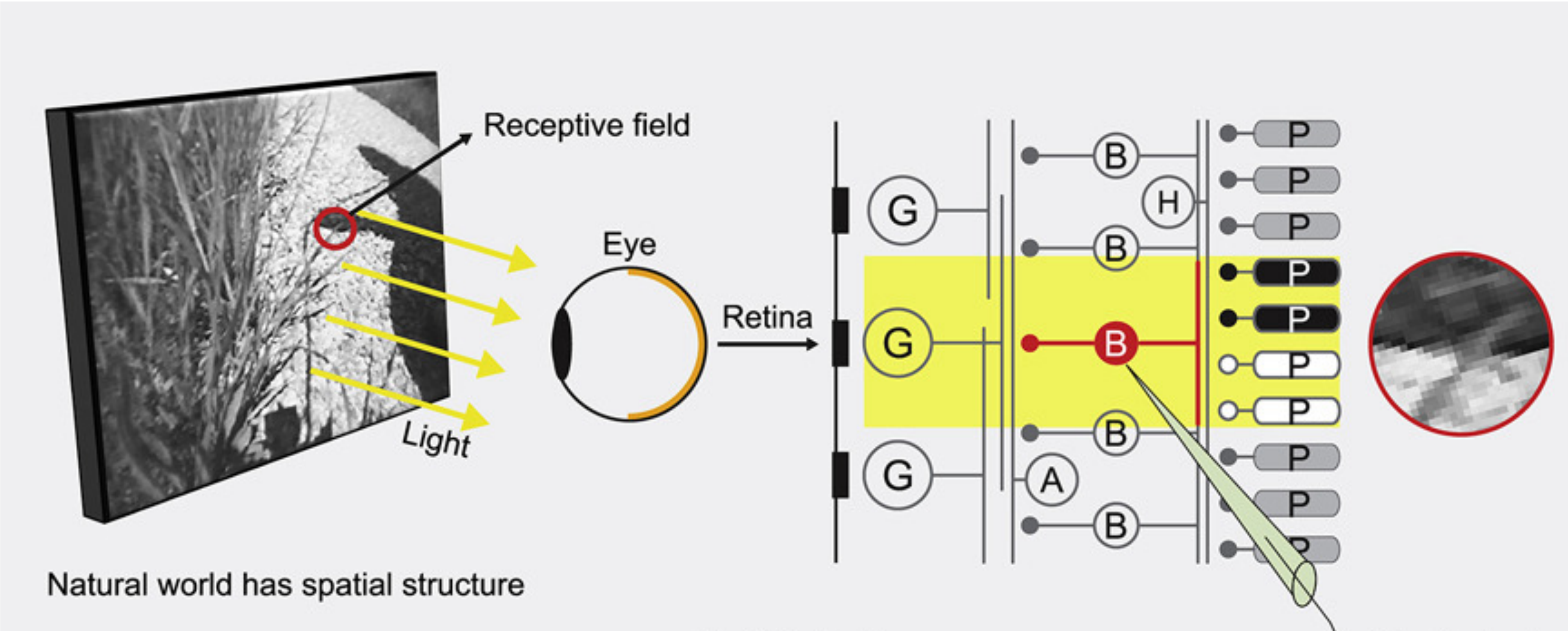
$$I = \int df \log_2[1 + S(f)/N(f)]$$

Efficient coding by maximizing mutual information under resource constraints (whiteboard)

Retina Circuit

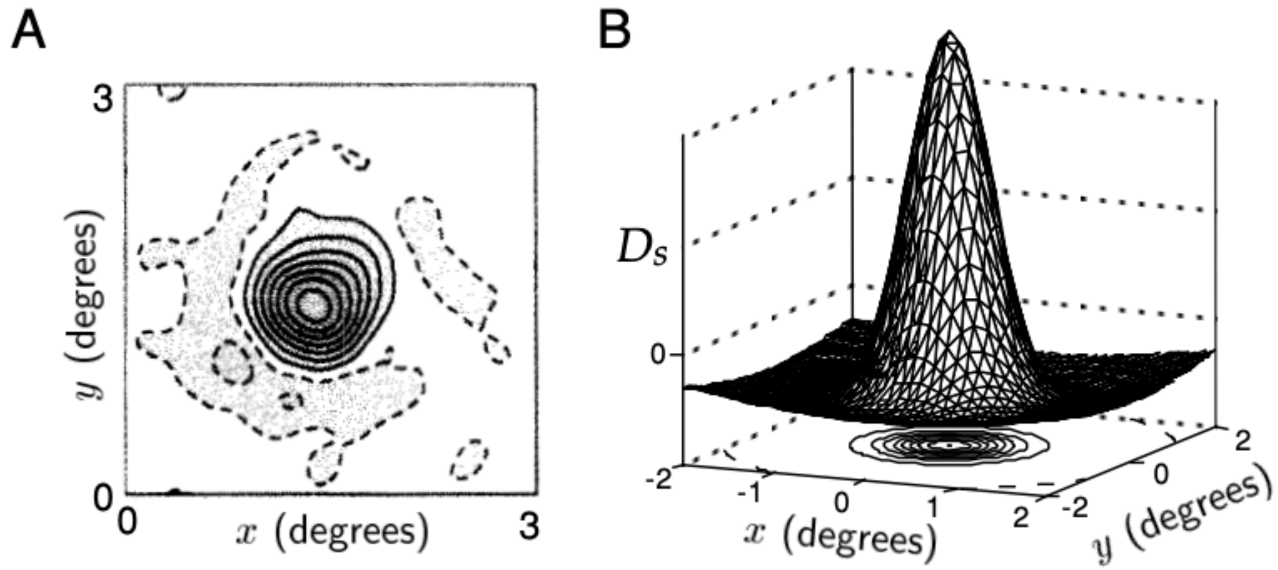


Retina Circuit



Schreyer & Gollisch 2021

Receptive field of retinal ganglion cells



Differences between Gaussians

$$D_s(x, y) = \pm\left(\frac{1}{2\pi\sigma_{\mathrm{cen}}^2}\exp\left(-\frac{x^2+y^2}{2\sigma_{\mathrm{cen}}^2}\right) - \frac{B}{2\pi\sigma_{\mathrm{sur}}^2}\exp\left(-\frac{x^2+y^2}{2\sigma_{\mathrm{sur}}^2}\right)\right).$$

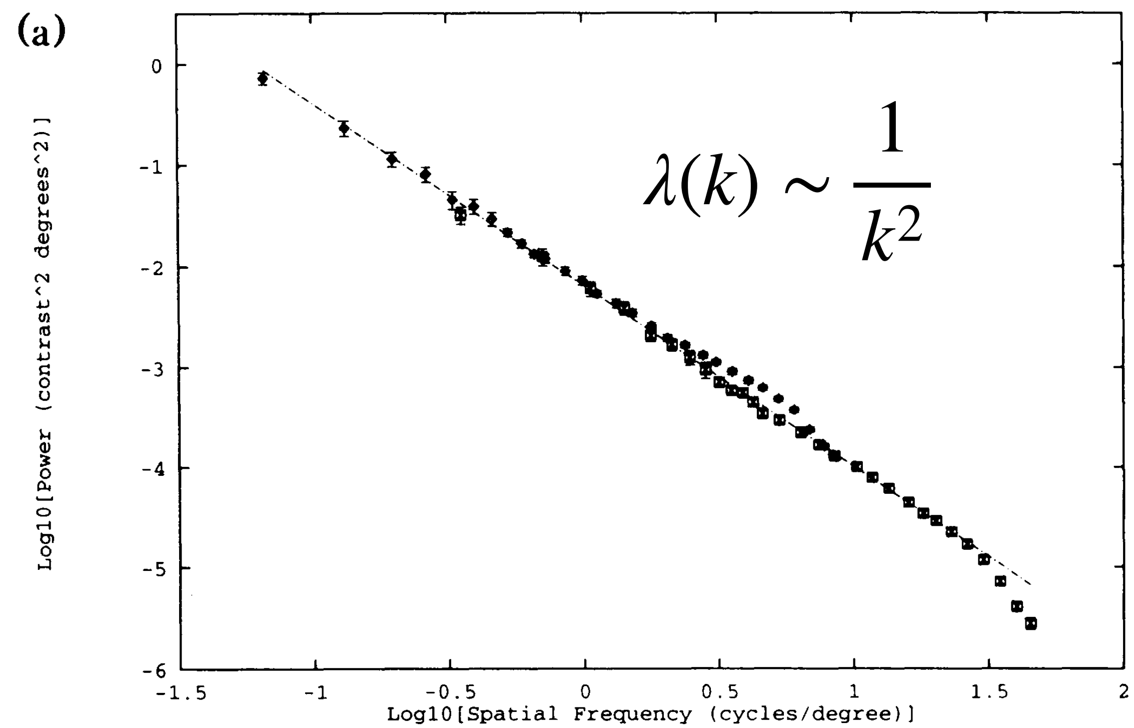
Statistics of inputs (Natural Scene)

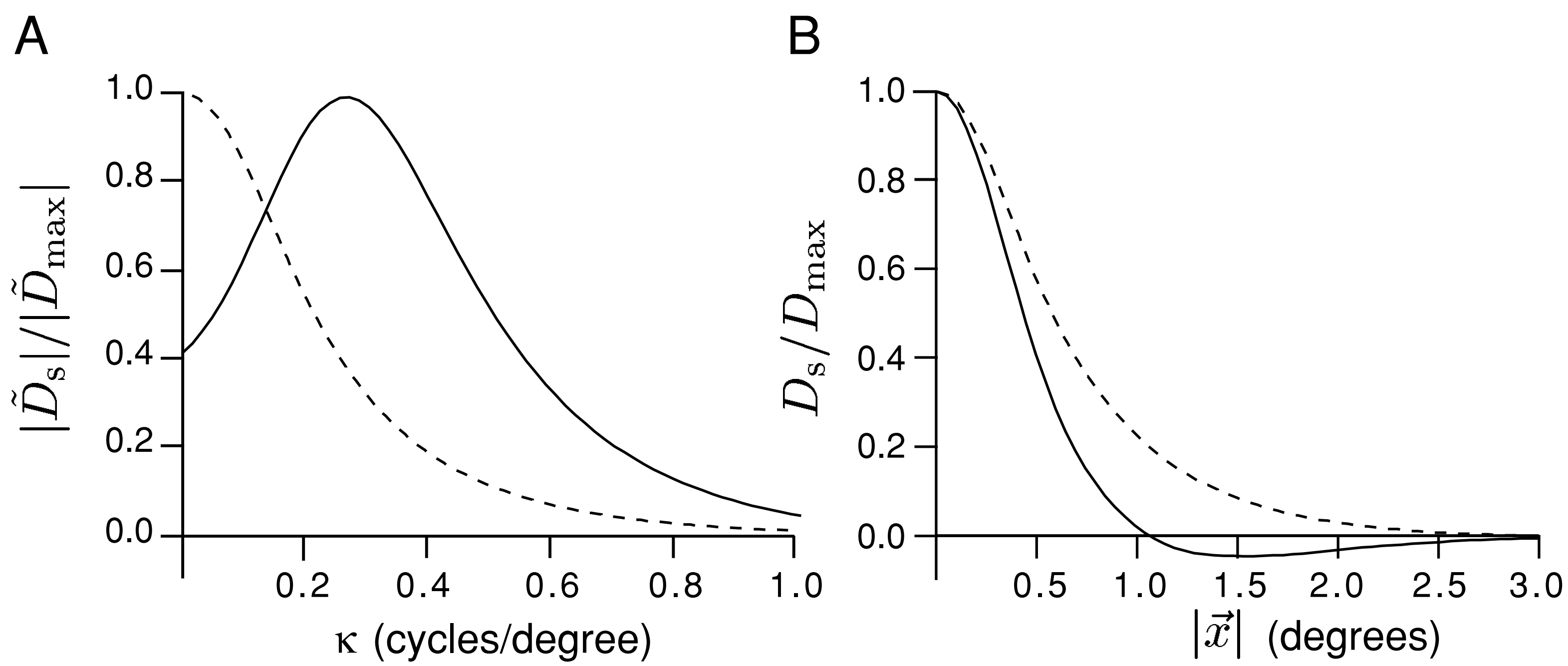
Covariance matrix of natural scenes is translationally invariant

$$\mathrm{cov}(I(x,y), I(x',y')) = f((x-x')^2 - (y-y')^2)$$

Eigenvectors of translationally-invariant matrices are Fourier modes.

Power spectrum density of natural scene is a power law




A
B
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.4
0.6
0.8
1.0
$|\tilde{D}_s|/|\tilde{D}_{max}|$
κ (cycles/degree)
D_s/D_{max}
0.5
1.0
1.5
2.0
2.5
3.0
$|\vec{x}|$ (degrees)

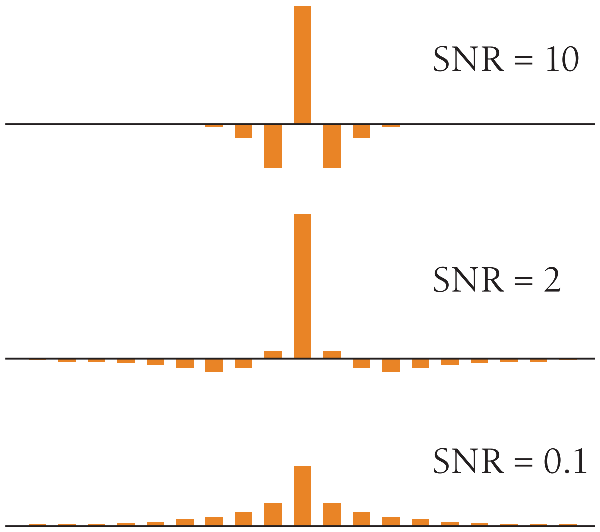


FIGURE 6.33

Cross-sections through the optimal matrices W_{ij} in the problem of efficient coding with noise. The correlation function is assumed to be exponential, $C_{ij} \propto \exp(-|i-j|/\xi)$, with $\xi = 50$, much longer than the range of interactions shown here. At high signal-to-noise ratios (SNRs), the solution looks like a differentiator, which removes the second-order correlations in the signal, whereas at low SNR, the solution integrates to suppress noise. Redrawn from Atick and Redlich (1990).

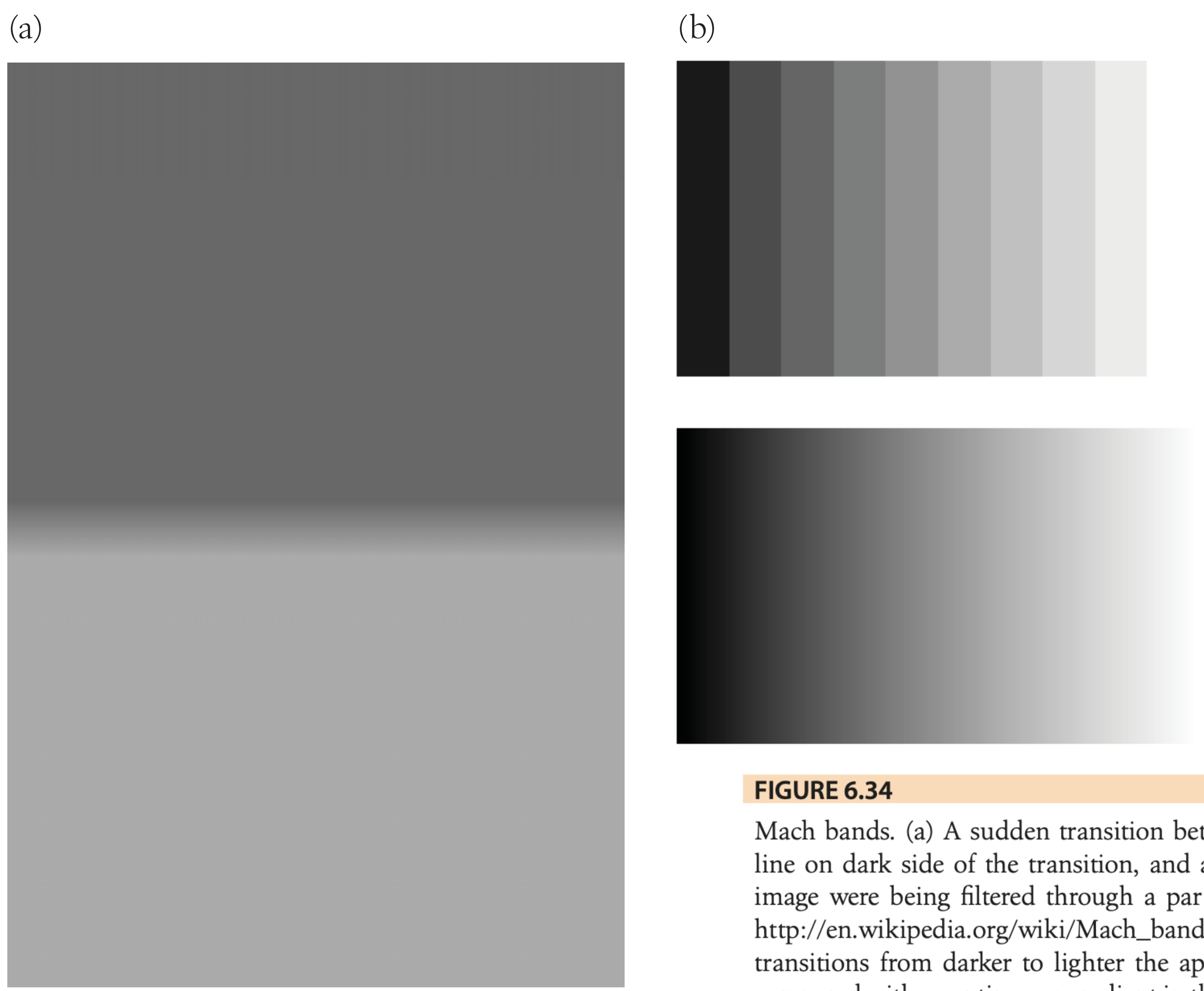
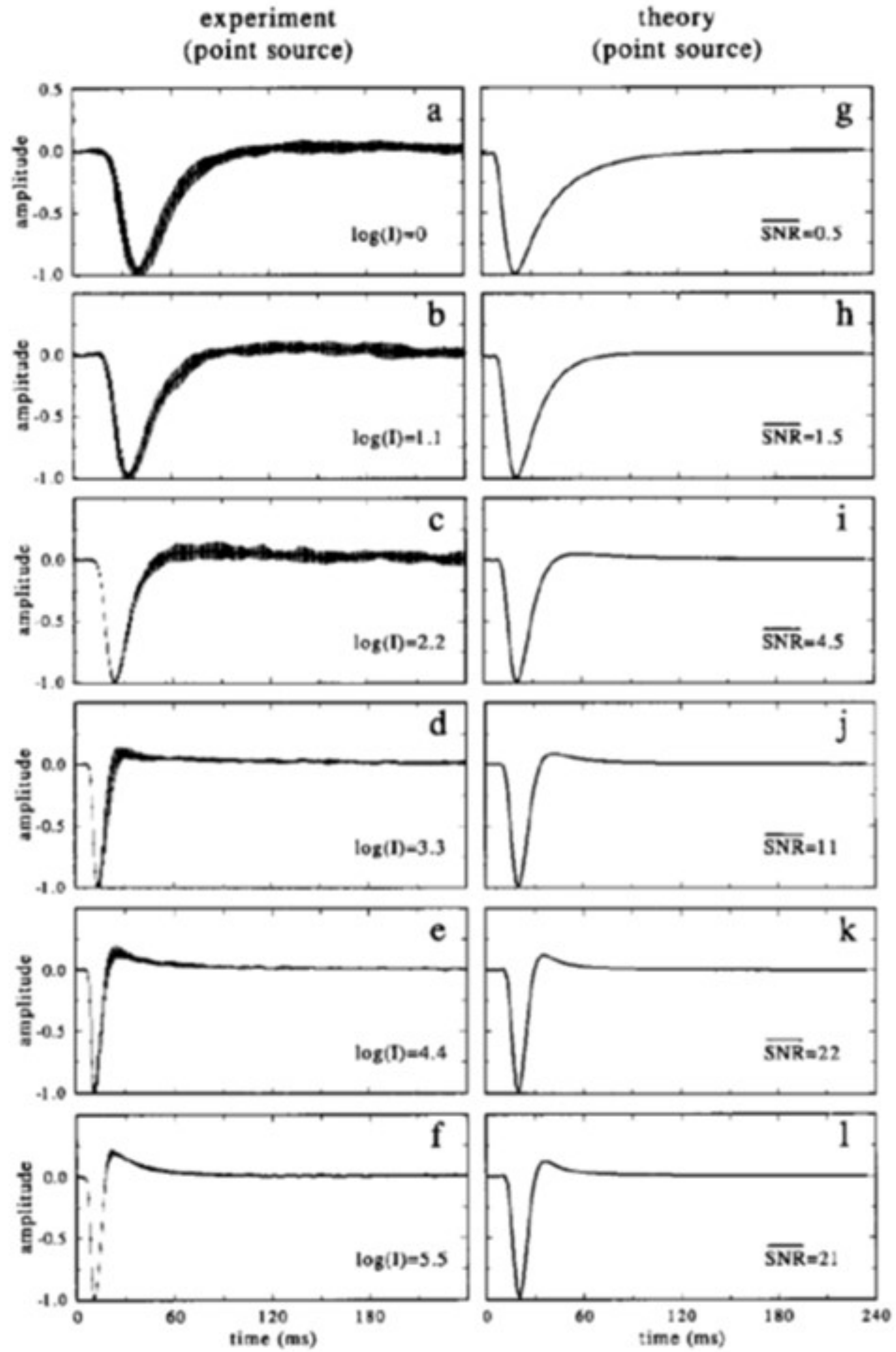


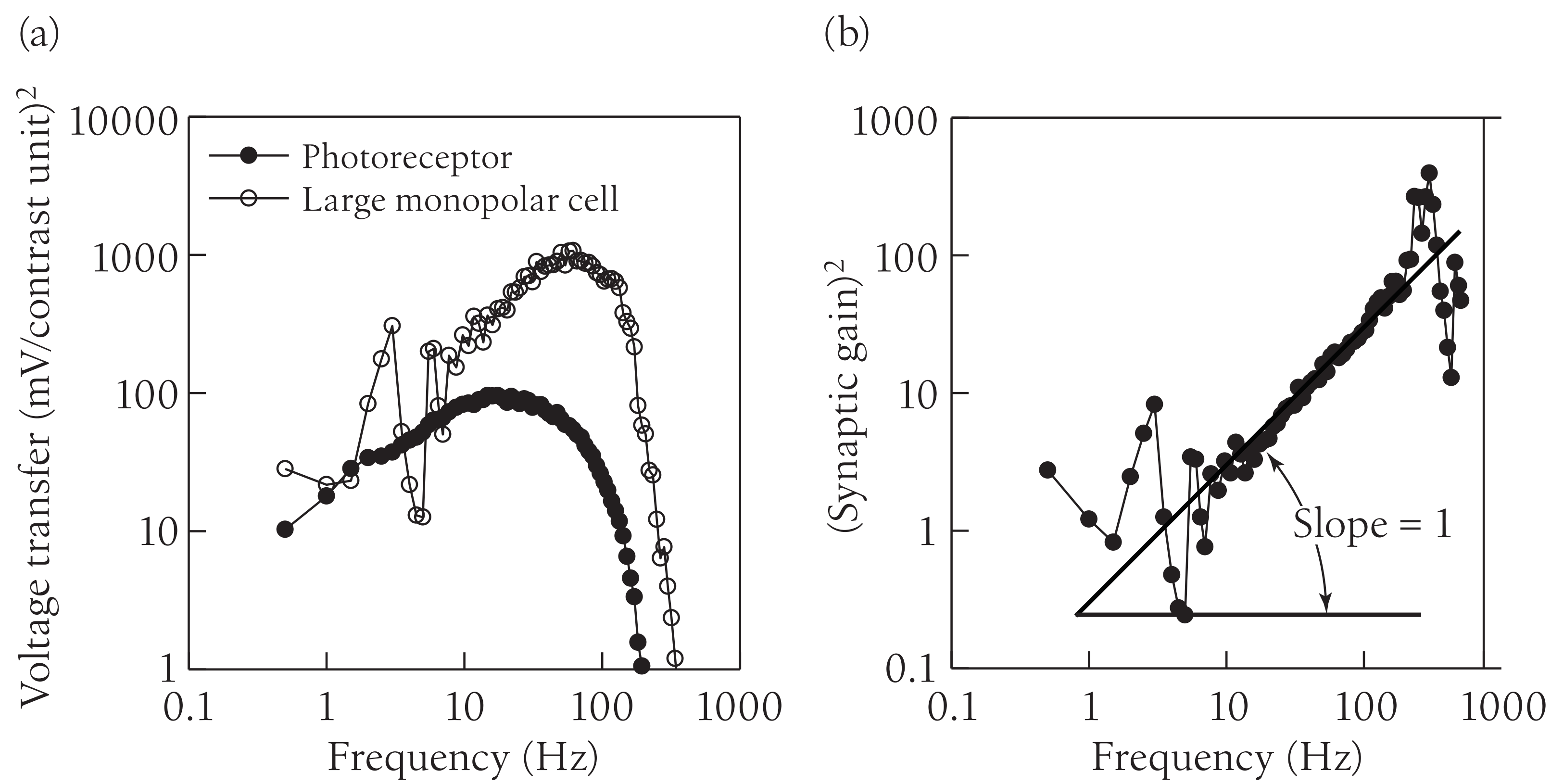
FIGURE 6.34

Mach bands. (a) A sudden transition between gray levels is perceived with an extra dark line on dark side of the transition, and an extra bright line on the bright side, as if the image were being filtered through a partially differentiating filter as in Fig. 6.33. From http://en.wikipedia.org/wiki/Mach_bands. (b) This filtering gives a sequence of discrete transitions from darker to lighter the appearance of nonmonotonicity, especially when compared with a continuous gradient in the bottom image. From J. Pomerantz, Rice University, with my thanks.



Fly Large Monopolar cell (LMC)

J H van Hateren 1992

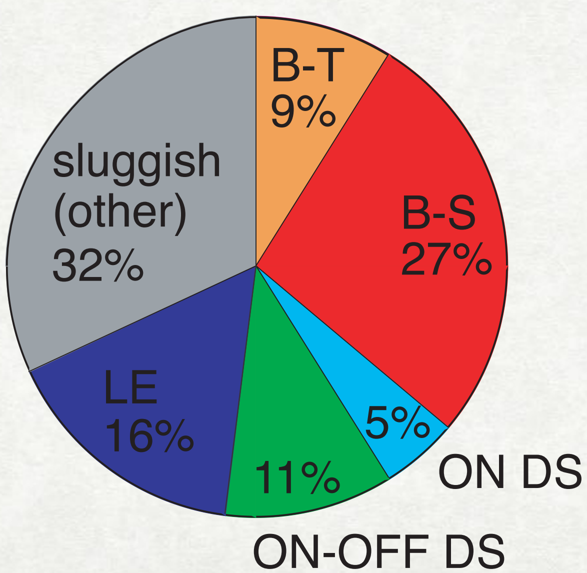

(a)
Voltage transfer (mV/contrast unit)^2
Photoreceptor
Large monopolar cell
10000
1000
100
10
1
0.1
1
10
100
1000
Frequency (Hz)
(b)
(Synaptic gain)^2
Slope = 1
1000
100
10
1
0.1
0.1
1
10
100
1000
Frequency (Hz)

A paradox

Information rate of the visual system

Information traffic along the optic nerve

cell type	bits/s	bits/spike	cells	bits	spikes
Brisk-transient	13	1.9	6,000	78,000	41,000
Brisk-sustained	10	1.8	24,000	240,000	133,000
ON DS	6	2.2	7,000	42,000	19,000
ON-OFF DS	8	2.2	12,000	96,000	44,000
Local-edge	7	2.1	20,000	140,000	67,000
Sluggish (other)	9	2.2	31,000	279,000	127,000
Total			**100,000**	**875,000**	**431,000**



- Messages are transmitted asymmetrically - less studied sluggish cells account for most of the traffic

- Scaling up to human optic nerve (million axons) gives a traffic of order an Ethernet cable. This is the same order of magnitude as the amount of information in natural scenes according to Ruderman, 1994.

Information rate of human behavior

Behavior/Activity	Time scale	Information rate (bits/s)	References
Binary digit memorization	5 min	4.9	[10]
Blindfolded speedcubing	12.78 sec	11.8	[7]
Choice-reaction experiment	min	~ 5	[126–128]
Listening comprehension (English)	min - hr	$\sim$13	[6]
Object recognition	1/2 sec	30 - 50	[129]
Optimal performance in laboratory motor tasks	$\sim$15 sec	10 - 12	[120, 121]
Reading (English)	min	28 - 45	[22]
Speech in 17 languages	< 1 min	39	[122]
Speed card	12.74 sec	17.7	[11]
StarCraft (e-athlete)	min	10	[119]
Tetris (Rank S)	min	< 6	[118]
Typing (English)	min - hr	10	[4, 5]

Table 1: The information rate of human behaviors

Typing (English): $$I = 2\frac{\text{words}}{\text{s}} \cdot 5\frac{\text{characters}}{\text{word}} \cdot 1\frac{\text{bit}}{\text{character}} = 10\frac{\text{bits}}{\text{s}}$$

Zheng & Meister 2024 https://arxiv.org/pdf/2408.10234v1

The premise of Brain-Machine Interface

"Based on the research reviewed here regarding the rate of human cognition, we predict that Musk's brain will communicate with the computer at about 10 bits/s. Instead of the bundle of Neuralink electrodes, Musk could just use a telephone, whose data rate has been designed to match human language, which in turn is matched to the speed of perception and cognition."

Zheng & Meister 2024 https://arxiv.org/pdf/2408.10234v1